

Programme	B. Sc. Mathematics Honours			
Course Code	MAT2MN102			
Course Title	CALCULUS AND MATRIX ALGEBRA			
Type of Course	MINOR			
Semester	II			
Academic Level	100-199			
Course Details	Credit	Lecture/Tutorial per week	Practicum per week	Total Hours
	4	4	-	60
Pre-requisites	Basic Calculus			
Course Summary	Students learn about antiderivatives, the indefinite and definite integrals, Riemann sums, and the Fundamental Theorem of Calculus. Course explores the average value of functions, evaluating definite integrals by substitution, calculating areas between curves, and finding the length of plane curves. Next it introduces functions of multiple variables, including notation, graphs, limits, continuity, and partial derivatives for functions of two or more variables. Course also focuses on matrix algebra, determinants, eigenvalue problems (including complex eigenvalues), and orthogonal matrices and their properties.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate proficiency in applying calculus techniques to solve analytical and geometrical problems involving indefinite and definite integrals, substitution methods, and integration by parts.	Ap	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	Apply multivariable calculus concepts, including functions of multiple variables, limits, continuity, and partial derivatives, to model and analyse real-world phenomena and mathematical problems.	Ap	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Apply linear algebra principles, such as matrix operations, determinants, and eigenvalue problems, to analyze and solve systems of equations and geometric problems.	Ap	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book	1. Howard Anton, Bivens and Stephen Davis, Calculus- Early Transcendentals (10 th Edition). 2. Advanced Engineering Mathematics(6/e): Dennis G Zill Jones & Bartlett, Learning, LLC (2018) ISBN: 9781284105902			
Module	Unit	Content	Hrs 60	External Marks (70)
I	Indefinite and Definite Integrals		12	Min 15
	1	Section 5.2: The Indefinite Integral - Antiderivatives, The Indefinite Integral, Integration Formulas, Properties of the Indefinite Integral, Integral Curves		
	2	Section 5.3: Integration by Substitution - u-Substitution, Easy to Recognize Substitutions, Less Apparent Substitutions		
	3	Section 5.5: The Definite Integral - Riemann Sums and the Definite Integral, Properties of the Definite Integral.		
	4	Section 5.6: The Fundamental Theorem of Calculus - The Fundamental Theorem of Calculus (sub section), The Relationship Between Definite and Indefinite Integrals.		
II	Techniques and Applications		13	Min 15
	5	Section 5.8: Average Value of a Function and its Applications - Average Value of a Continuous Function (up to and including Example 2 only)		
	6	Section 5.9: Evaluating Definite Integrals by Substitution - Two Methods for Making Substitutions in Definite Integrals		
	7	Section 6.1: Area Between Two Curves - Area Between $y = f(x)$ and $y = g(x)$, Reversing the Roles of x and y		
	8	Section 6.4: Length of a Plane Curve - Arc Length		
	9	Section 7.2: Integration by Parts - The Product rule and Integration by Parts, Guidelines for Integration by Parts, Repeated Integration by Parts		
	10	Section 7.5: Integrating Rational Functions by Partial Fractions - Partial Fractions, Finding the form of a Partial Fraction Decomposition, Linear Factors, Quadratic Factors (Example 4 is optional), Integrating Improper Rational Functions.		
III	Multivariable Calculus		10	Min 15
	11	Section 13.1: Functions of Two or More Variables: Notation and Terminology, Graphs of Functions of Two Variables.		
	12	Section 13.1: Functions of Two or More Variables: Level Curves, Level Surfaces.		
	13	Section 13.2: Limits and Continuity - Limit along Curves		
	14	Section 13.2: Limits Continuity - Continuity		
	15	Section 13.3: Partial Derivatives -		

		Partial Derivatives of Functions of Two Variables, The Partial Derivative Function, Partial Derivative Notation, Implicit Partial Differentiation, Partial Derivatives and Continuity		
	16	Section 13.3: Partial Derivatives Partial Derivatives of Functions with more than Two Variables, Higher order Partial Derivatives, Equality of Mixed Partials.		
IV	Linear Algebra Essentials		13	Min 15
	17	Section 8.1: Matrix Algebra		
	18	Section 8.2: Systems of Linear Algebraic Equations		
	19	Section 8.8: The Eigenvalue Problem - Topics up to and including Example 4		
	20	Section 8.8: The Eigenvalue Problem - Topics from Complex Eigenvalues onwards		
	21	Section 8.10: Orthogonal Matrices - Topics up to and including Theorem 8.10.3		
	22	Section 8.10: Orthogonal Matrices - Topics from Constructing an Orthogonal Matrix onwards		
V	Module V (Open Ended)		12	
		Fundamental theorems in Vector Calculus such as Green's theorem, divergence theorem, and the Stokes' theorem.		
		Trigonometric Substitutions		
		Integrating Trigonometric Functions		
		Volume of Solids of Revolution, Area of Surfaces of Revolution		
		The Chain Rule in Partial Differentiation		
		Directional Derivatives and Gradients, Tangent Planes and Normal Vectors		
		Basics of Vector Calculus including the differential operators such as gradient, divergence and curl.		
		Simpsons Rule, Trapezoidal rule in Numerical Integration		
		Algebra of Complex Numbers		
References				
	1	Calculus and Analytic Geometry, 9 th Edition, George B. Thomas Jr and Ross L. Finney, Pearson Publications.		
	2	Calculus, Soo T. Tan, Brooks/Cole Cengage Learning (2010) ISBN-13: 978-0-534-46579-7.		
	3	Marsden, Jerrold, and Alan Weinstein. <i>Calculus I</i> . Springer Science & Business Media, 1985.		
	4	Stein, Sherman K. <i>Calculus in the first three dimensions</i> . Courier Dover Publications, 2016.		
	5	Kreyszig, Erwin. <i>Advanced Engineering Mathematics 9th Edition with Wiley Plus Set</i> . Vol. 334. US: John Wiley & Sons, 2007.		
	6	Elementary Linear Algebra, Applications version, 9 th edition, Howard Anton and Chriss Rorres		

Note: 1) Optional topics are exempted for end semester examination. 2) Proofs of all the results are also exempted for the end semester exam.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	3	1	1	1	3	0	0
CO 2	2	1	2	1	2	1	2	0	0
CO 3	2	1	2	1	2	1	2	0	0

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓